Dr. K. Balamurugan
Senior Professor & Head

Contact

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Academic Qualifications:

M. Sc., Ph. D., D. Sc.,

Teaching Experience:

21 Years

Research Experience:

27 Years

Additional Responsibilities

- Coordinator: RUSA 2.0 Scheme, Alagappa University (Since 23 Nov 2022-)
- Member of the Institutional Animal Ethical Committee (IAEC)
- Director: IPR Cell, Alagappa University (Since 09 Dec 2022-)
- Member Secretary, Institutional Ethical Committee (Since March 2023-)
- Coordinator: Incubation and Technology Transfer center (Govt. of TN Scheme) (01-05-2015 to 19 August 2019)
- Chief Superintendent –PG Examinations- November 2018 and April 2019 @ Science Campus, Alagappa University

- Special Officer for Recruitment Cell of Alagappa University [24 October 2018 to 11 February 2020]
- Expert Member of the Inspection Commission for Affiliation of B.Sc., and M. Sc., Biotechnology
- Co-Coordinator: Bioinformatics Infrastructure Facility (funded by DBT, Govt. of India)
- Deputy Coordinator: UGC-SAP (DRS-I) Department of Biotechnology.
- RUSA 2.0 Biological Theme Coordinator (Sep 2019- Mar 2023)

Areas of Research

- **Host pathogen-interactions using *C. elegans* as model organism**
- **Understanding innate immune regulations through functional genomics, proteomics and metabolomics**
- **Studying the multigenerational adaptation during bacterial infections through multi-OMICS study**

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	**PDF**	01	-
	Ph.D.	14	06
	M.Phil.	-	-
Project	**PG**	69	04
	UG / Others	12	-

Publications

International		National		Others
Journals	**Conferences**	**Journals**	**Conferences**	**Books / Chapters / Monographs / Manuals**
112	170	04	64	19

Cumulative Impact Factor (as per JCR) : 362.532 (Avg. I.F. 3.12)
h-index : 31
i10 index : 72
Total Citations : 2528

Funded Research Projects

Completed Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1.	DBT	Dec-2007	Dec-2010	RNA-interference mediated silencing of antimicrobial genes of *Caenorhabditis elegans.*	24.73
2.	UGC	Apr-2008	Mar-2011	Screening of marine bioresources for antibacterial compounds	11.76
3.	ICMR	Mar-2011	Mar-2014	Analysis of antimicrobial gene expression pattern	25.23
4.	DBT	Mar-2011	Mar-2014	Studies on Immune regulatory proteins	26.546
5.	CSIR	Apr-2011	Mar-2014	*Caenorhabditis elegans* response to human pathogens	21.92
6.	DST	July-2011	July-2014	Characterization of innate immune regulators during *Shigella spp.*	23.30
7.	DST	Feb-2012	Jan-2015	*Caenorhabditis elegans* response against *Vibrio spp.* infection	20.03
8.	ITC-AU Collaborative project	July-2011	Dec-2014	Anti-aging: Role of target genes	19.14
9.	UGC Major Research Project	01 April 2013	31 March 2017	Physiological and molecular changes in *Caenorhabditis elegans* during subsequent bacterial infections	10.85
10.	ITC-AU Collaborative Project	Jul-2016	Dec-2019	*C. elegans*: An *in vivo* model for dermal inflammation studies	41.17
11.	DBT	June 2017	June 2020	Impact of *Cronobacter sakazakii* infection on the neuroimmunity	70.24

On-going Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1	ITC-AU Collaborative Project	05 Oct-2022	04 Oct-2023	Identification and use of antiviral natural actives using *in silico*, in vitro and in vivo models	39.60
2	DST-SERB	20 Dec 2021	19 Dec	Impact of specific metabolites during bacterial infection and their role in	39.46

2024 multigenerational adaptation

Patents

S. No.	Title	Inventors	Patent Number	Filing Date	Publication date
1	Personal Care Compositions for Anti-Aging	Prasanth MI, **Balamurugan K**, Pandian SK, Gayathri S, James PB	676/KOL/2015	18 June 2015	20/10/17
		Granted on 03 June 2020; Patent Number: 337898			
2	Personal Care Compositions for Anti-Aging	Prasanth MI, **Balamurugan K**, Pandian SK, Gayathri S, James PB	679/KOL/2015	18 June 2015	20/10/17
3	Composition Comprising Green Tea and Naringenin for Anti-Aging	Prasanth MI, **Balamurugan K**, Pandian SK, Gayathri S, James PB	677/KOL/2015	18 June 2015	20/10/17
4	Personal Care Compositions for Anti-Aging	Prasanth MI, **Balamurugan K**, Pandian SK, Gayathri S, James PB	680/KOL/2015	18 June 2015	20/10/17
5	Personal Care Compositions for Anti-Aging	Prasanth MI, Pandian SK, Gayathri S, James PB, **Balamurugan K**	766/KOL/2015	15 July 2015	13/10/17
6	Personal Care Compositions for Anti-Aging	Prasanth MI, Pandian SK, Gayathri S, James PB, **Balamurugan K**	779/KOL/2015	17 July 2015	13/10/ 17
7	An anti-acne synergistic composition and process thereof	Sivasankar C, Pandian SK and **Balamurugan K**	Appln. No. 201641010057	22 March 2016	29/09/2017
		Granted on 03.11.21; Patent No. 381269			
8	An antibacterial composition and implementations thereof	Swetha TK, Pandian SK, Sivasankar C, **Balamurugan K**, Veera Ravi A, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008480	07 March 2018	13/09/2019
9	A composition comprising phytochemicals and applications thereof	Swetha TK, Pandian SK, Sivasankar C, **Balamurugan K**, Veera Ravi A, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008481	07 March 2018	13/09/2019
10	A composition comprising phytochemicals and applications thereof	Swetha TK, Pandian SK, Sivasankar C, **Balamurugan K,** Veera Ravi A, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008482	07 March 2018	13/09/2019

11	Antibacterial composition and uses thereof	Swetha TK, Pandian SK, Sivasankar C, **Balamurugan K**, Veera Ravi A, Bhaskar JP, Venkateswaran K, Deepa M, Das SS	Application No. 201831008483	07 March 2018	13/09/2019
12	AN ANTI-AGING COMPOSITION	Muthubharthi B. C., Mounish B.S.C., Subalakshmi K. P and **Balamurugan K.**	Application no. 202341021225Ref. No. PD047148IN-SC	24 March, 2023	Submitted
13	A NUTRITIONAL COMPOSITION AND IMPLEMENTATIONS THEREOF	Muthubharthi B. C., Mounish B.S.C. and **Balamurugan K.**	Application no. 202341021224Ref. No. PD047149IN-SC	24 March, 2023	Submitted
14	A COMPOSITION FOR ANTI-AGING AND PROCESS FOR THE PREPARATION OF THE COMPOSITON THEREOF	Muthubharthi B. C. and **Balamurugan K.**	Application no. 202341021222Ref. No. PD047150IN-SC	24 March, 2023	Submitted
15	AN AGE DEFYING COMPOSITION	Muthubharthi B. C. and **Balamurugan K**.	Application no. 202341021223Ref. No. PD047151IN-SC	24 March, 2023	Submitted
16	A SUPPLEMENT COMPOSITION	Muthubharthi B. C. and **Balamurugan K**.	Application no. 202341021221Ref. No. PD047153IN-SC	24 March, 2023	Submitted

Distinctive Achievements / Awards

1. Vallal Alagappar Research Recognition Award-2020- contribution towards the enhancement of Research outcome of the University in the form of "h" index
2. **TamilNadu Scientist Award (TANSA-2020) under the category of Biological Sciences**
3. Outstanding Researcher Award-2022 in recognition of the contribution made towards Excellence in Research at Alagappa University
4. 2008-DBT-RGYI Young Scientist award project for Young Investigator under 40 Years.

5. 2011- DST Young Scientist Award project.
6. 2011-DST International Travel award for attending FEMS 2011 Conference held at Geneva, Switzerland during June 26-30, 2011.
7. Best Poster Award -"**AMI-Panjab University**" for the poster presentation entitled "*Modification of pathogen lipopolysaccharide during the interaction with Caenorhabditis elegans*" presented by Vignesh Kumar B and Balamurugan, K, at 52nd Annual Conference of Association of Microbiologists of India (AMI), "International Conference on Microbial Biotechnology for Sustainable Development" during Nov 3-6, 2011 held at Panjab University, Chandigarh, India.
8. Invited ORAL presentation at the 10th Asia Pacific Bioinformatics Conference conducted by LA Trobe University, Melbourne, Australia, 17-19 Jan 2012.
9. Second Best Oral Presentation Award at the 5th International Conference on Natural Products for Health and Beauty (NATPRO 5) in Phuket, Thailand, 05-07 May 2014.
10. **Best Faculty Award- Biotechnology 2013- Senior** by Shri PK Das Memorial, Nehru Group of Institutions, Coimbatore.
11. **Dr. R. R. Mani Maran Memorial Lecture Award** on 26 November 2014 by Indian Society for Comparative Endocrinology for the scientific contribution in the field of Host-Pathogen interactions related to Reproduction.
12. Adjunct Faculty, National Institute of Pharmaceutical Education and Research (NIPER) Kolkata, since 06 Dec 2016
13. Visiting Professor, Department of Biochemistry and Molecular Biology, Faculty of Agriculture and Life Science, Hirosaki University, Japan, March 12, 2018- March 28, 2018.
14. Pooranachithra M., Balamurugan K. "Wound Research: Insights from a Mighty Miniature Model, *Caenorhabditis elegans*". **Augmenting Writing Skills for** Articulating Research (AWSAR) Award 2018 on Feb, 28 2019 by Department of Science and Technology (DST), Government of India.
15. Subhadra M. and Balamurugan K. "A Tug of War between Diabetes and Obesity – Just a Glimpse" Augmenting Writing Skills for Articulating Research (AWSAR) Award 2020 Won the CONSOLATION PRIZE (Ph.D.) in the Popular Science by DST, GOI.
16. Best Poster Award at the "Workshop for Agriculture and Life science study in Hirosaki University 2021- models for Life science studies" organized by Hirosaki University, Japan held during January 23rd-24th, 2021. (Poster no: 05-Virtual Conference); presented by Muthubharathi B C and Balamurugan K., on "Understanding the importance of Neuronal system during *Cronobacter sakazakii* infection in the model host, *Caenorhabditis elegans*.
17. Certificate of Appreciation-2021 as Theme Coordinator in Translational Health Research for Human Animal and Plant System in recognition of the contribution made to THEME BASED RESEARCH PROJECT OF RUSA 2.0 towards the quality enhancement in teaching and research by Alagappa University, Karaikudi, Tamil Nadu, India

Events organized in leading roles

Number of Seminars / Conferences / Workshops / Events organized: **09**

Organized an International Conference cum workshop on "*Caenorhabditis elegans* based OMICS for Future Challenges" during 09-13 September 2019

Organized Proteomics Day Celebration and Education Day on "Advances in Microbial and Clinical Proteomics" on March 18, 2021.

Organized Proteomics Day Celebration and Education Day on "Advances in Proteome Analysis" on March 21, 2022.

Organized "National Level Conference on Biotechnology and OMICS Sciences-2022" during May 27, 2022 – May 28, 2022.

Events Participated (optional)

Conferences / Seminars / Workshops: 189

Overseas Exposure / Visits

1. Singapore
2. Japan
3. Malaysia
4. Israel
5. Portugal
6. Thailand
7. Germany
8. London, United Kingdom
9. Australia
10. Switzerland
11. USA
12. Taiwan

Membership in

Professional Bodies

LIFE MEMBER: Association of Microbiologists of India
LIFE MEMBER: Society of Biological Chemists, India (SBC)
LIFE MEMBER: The Biotech Research Society, India (BRSI)
LIFE MEMBER: Indian Society for Comparative Endocrinology (ISCE)
LIFE MEMBER: Proteomics Society, India (PSI)

- The Indian Science Congress Association
- American Gastroenterological Association
- European Congress of Clinical Microbiology and Infectious Diseases
- American Society for Microbiology

Editorial Board

- Gene Reports

- Journal of Proteins and Proteomics
- CRC Press: Taylor and Francis

Advisory Board

- UGC-SAP (DRS-1) Advisory Committee member- Department of Biosciences, Mangalore University (2016-)
- Member of the Academic Council of Karpagam Academy of Higher Education (2018-19 & 2019-20)
- Senate member- NIPER-Kolkata (since 06 September 2019-)

Academic Bodies (such as Board of Studies etc.,)

MEMBER IN BOARD OF STUDIES

- M.Sc. Biotechnology, Alagappa University (2007 onwards)
- Biotechnology, Manonmaniam Sundaranar University, Tirunelveli (2008-2010)
- Biochemistry, Dr. G.R. Damodaran College of Science (2010-2012)
- B.Sc. (Chairman) Biotechnology, Alagappa University (2008-2011)&(2015-2017)
- B.Sc. Biotechnology, Alagappa University (2011-2014)
- B.Sc. Biochemistry, Alagappa University (2011-2014)
- B.Sc. Advanced Zoology & Animal Biotechnology (2015-2017)

Resource persons in various capacities

Number of Invited / Special Lectures delivered: **98**

Others

1. Articles published in Newspapers / Magazines : 16 **Book chapters**
2. Products developed : 0**7**
3. No. of PhD Thesis evaluated : **32**
4. No. of PhD Public Viva Voce Examination conducted : **24**
5. Sequences submitted in GenBank: **103**

Recent Publications (selective)

1.	Muthubharathi, B. C., Ravichandiran, V. and **Balamurugan, K** (2023). Distinct global metabolomic profile of the model organism Caenorhabditis elegans during the interaction of Staphylococcus aureus and Salmonella enterica Serovar Typhi. Molecular Omics [Publisher: Royal Society of Chemistry, London] Impact factor: 4.212. MO-RES-02-2023-000040.R1 (Accepted).
2.	Gowripriya T, Yashwanth R, Suresh R, James PB and **Balamurugan K.** (2023). A systematic review on impact of SARS-CoV-2 infection. *Microbiological Research.* Vol. 271: 127364; ISSN 0944-5013; https://doi.org/10.1016/j.micres.2023.127364.[Publisher: Elsevier, Netherlands]; [IF: 5.07].
3.	Muthubharathi, B. C., Ravichandiran, V. and **Balamurugan, K**. (2022). *Cronobacter sakazakii* cue for the attraction and its impact on the immunity of *Caenorhabditis elegans*. ASM Infection and Immunity. DOI: https://doi.org/10.1128/iai.00281-22. Vol 90, Issue 11, e00281-22 [IF: 3.441].
4.	Muthubharathi BC, Balasubramaniam B, Mir DA, Ravichandiran V, and **Balamurugan K** (2021). Physiological and metabolite alterations associated with neuronal signals of *Caenorhabditis elegans*

	during *Cronobacter sakazakii* infections. ACS Chemical Neuroscience. Doi: 10.1021/acschemneuro.1c00559 [Publisher: American Chemical Society, United States] [IF: 4.418]
5.	Prithika U and **Balamurugan K** (2021). Evaluation of changes in *C. elegans* immune response during bacterial infection: A single nematode approach. *Microbes and Infection* Doi: 10.1016/j.micinf.2021.104846 [ELSEVIER; France] (Impact Factor: 2.373).
6.	Mir DA, Balasubramaniam B, VenkataKrishna L, Muthubharathi BC, and **Balamurugan K** (2021). Proteomic analysis of *Caenorhabditis elegans* against *Salmonella Typhi* toxic proteins. *Genes and Immunity.* doi: 10.1038/s41435-021-00132-w. [Publisher: Springer Nature] [IF: 4.481]
7.	Pooranachithra, M., Suruthi, K., Prabhanand, B. J., Deepa, M., Sekhar, D. S., Venkateswaran, K., Rahul, G., Ravichandiran V., **Balamurugan, K.** (2021) Survival upon *Staphylococcus aureus* mediated wound infection in *Caenorhabditis elegans* and the mechanism entailed. *Microbial Pathogenesis*. Aug;157:104952. doi: 10.1016/j.micpath.2021.104952. (Impact factor: 2.914).
8.	Pooranachithra M, SatheeshKumar C, Bhaskar JP, Venkateswaran K, Ravichandiran V, **Balamurugan K** (2021) Proteomic analysis of *Caenorhabditis elegans* wound model reveals novel molecular players involved in repair. *Journal of Proteomics* 240 (2021): 104222 DOI: 10.1016/j.jprot.2021.104222. [Elsevier, Country: UK] [IF: 3.509].
9.	Muthubharathi BC, Gowripriya T. and **Balamurugan K** (2021). Metabolomics-The small molecules matter more. *Molecular Omics* 17(2):210-229. doi: 10.1039/d0mo00176g. [Publisher: Royal Society of Chemistry, London] [IF: 2.273].
10.	Pooranachithra M, Bhaskar JP, **Balamurugan K** (2021). Wounding *Caenorhabditis elegans* with Glass Wool. *Bio Protocol.* Jan 20;11(2):e3885.[Publisher: Online by Standford, USA] DOI: 10.21769/BioProtoc.3885.
11.	Balasubramaniam, B., Gowripriya T., and **Balamurugan K** (2020). Identification of differentially regulated proteins of *Caenorhabditis elegans* during *Salmonella enterica* Serovar Typhi exposure using Mass Spectrometry. *Journal of Proteins and Proteomics.* [Publisher: Springer Nature, Switzerland] DOI: 10.1007/s42485-020-00033-y.
12.	Dilawar AM and **Balamurugan K** (2020). Modulation of the host cell mitochondrial proteome by PemKSa toxin protein exposure. Microbial Pathogenesis 140: p103963 [Publisher: Elsevier; Country UK; Impact Factor- 2.581].
13.	Prasanth MI, Gayathri S, Bhaskar JP, Krishnan V, **Balamurugan K** (2020). Understanding the role of p38 and JNK mediated MAPK pathway in response to UV-A induced photoaging in *Caenorhabditis elegans. J Photochem Photobiol B.* DOI: 10.1016/j.jphotobiol.2020.111844 [In Press] (Impact Factor: 4.067).
14.	Balasubramaniam, B., VenkataKrishna, L.M., Vinitha, T., JebaMercy, G., and **Balamurugan, K** (2020). *Salmonella enterica* Serovar Typhi exposure elicits deliberate physiological alterations and triggers the involvement of ubiquitin mediated proteolysis pathway in *Caenorhabditis elegans*. *International Journal of Biological Macromolecules* Vol. 149, Jan 24; 2020;149:215-233. (Impact Factor: 4.784). DOI: 10.1016/j.ijbiomac. 2020.01.225 [Publisher: Elsevier BV] [IF: 4.784].
15.	Prasanth MI, Gayathri S, Bhaskar JP, Krishnan V and **Balamurugan K** (2020). Analyzing the individual and synergistic effects of antioxidants in combating aging and photoaging using model nematode, *Caenorhabditis elegans*. Photochemistry and Photobiology 96 (1): 139-147 [DOI: 10.1111/php.13167]; [Publisher: Wiley-Blackwell, Inc.] [IF: 2.338].
16.	Dilawar AM and **Balamurugan K** (2019). *In vitro* and *in vivo* efficacy of *Caenorhabditis elegans* recombinant antimicrobial peptide against Gram-negative bacteria. Biofouling. Sep;35(8):900-921. doi: 10.1080/08927014.2019.1675048. [Publisher: Taylor & Francis, UK]; (Impact Factor: 2.847).
17.	Rai P, Sharika R, Ganguli A, **Balamurugan K**, Sarala B, Sharma R, Gupta R, Neogi SB (2019). Indigenous preparations of Bryonia laciniosa, Quercus infectoria, Putranjiva roxburghii and Mesua ferrea induces developmental toxicity in *C. elegans.* Proceedings of the National Academy of Sciences, India Section B: Biological Sciences [DOI: 10.1007/s40011-019-01138-1] [Publisher: Springer India; IF. 0.396].

18.	Pooranachithra M, Bhaskar JP, Murali D, Das SS, JebaMercy G, Krishnan V, **Balamurugan K**. (2019). Unravelling the wound healing ability and mode of action of pyridine carboxamide oxime using *Caenorhabditis elegans* as potential prescreen wound model. Life Sciences 235:116859; [DOI: 10.1016/j.lfs.2019.116859]; [Publisher: Elsevier BV] [I.F. 3.448].
19.	Balasubramaniam B, Vinitha T, Deepika S, JebaMercy G, VenkataKrishna LM and **Balamurugan K** (2019). Analysis of *Caenorhabditis elegans* phosphoproteome reveals the involvement of a molecular chaperone, HSP-90 protein during *Salmonella enterica* Serovar Typhi infection. International Journal of Biological Macromolecules 137:620-646. https://doi.org/10.1016/j.ijbiomac.2019.06.085; [Publisher: Elsevier BV] [IF: 4.784].
20.	Dilawar AM and **Balamurugan K** (2019). A Proteomic analysis of *Caenorhabditis elegans* mitochondria during bacterial infection. Mitochondrion. pii: S1567-7249(18)30138-7; DOI: 10.1016/j.mito.2019.03.002;. [Publisher: Elsevier BV; Impact Factor: 3.449].
21.	Dilawar AM and **Balamurugan K**. (2019). Global proteomic response of Caenorhabditis elegans against PemKSa toxin. Frontiers in Cellular and Infection Microbiology, section Bacteria and Host: 9:172; DOI: 10.3389/fcimb.2019.00172 [IF. 3.518].
22.	Kalaiyarasan G, Veerapandian M, JebaMercy G, **Balamurugan K**, Joseph J (2019). Amygdalin-Functionalized Carbon Quantum Dots for Probing β Glucosidase Activity for Cancer Diagnosis and Therapeutics. ACS Biomaterials Science & Engineering 2019 [DOI: 10.1021/acsbiomaterials.9b00394]; [I.F. 4.432]
23.	Pandey G, Marimuthu M, Kanagavalli P, Ravichandiran V, **Balamurugan K**, Veerapandian M (2019). Chitosanylated MoO3-Ruthenium(II) Nanocomposite as Biocompatible Probe for Bioimaging and Herbaceutical Detection. ACS Biomaterials Science & Engineering. (DOI: 10.1021/acsbiomaterials.9b00575) [I.F. 4.432]
24.	Prasanth MI, Venkatesh D, Murali D, Bhaskar JP, Krishnan V and **Balamurugan K** (2019). Understanding the role of DAF-16 mediated pathway in *Caenorhabditis elegans* during UV-A mediated photoaging process. *Archives of Gerontology and Geriatrics* [Publisher: ELSEVIER BV, Country: Netherlands; Impact Factor: 2.241].
25.	Mohana M, PraveenKumar B, Mariya Salomi L, Murugan V and **Balamurugan K** (2018). Methylene Blue-Fortified Molybdenum Trioxide Nanoparticles: Harnessing Radical Scavenging Property. ACS Applied Materials & Interfaces, Dec 19;10(50):43429-43438. doi: 10.1021/acsami.8b15841. [Publisher: American Chemical Society (United States); Impact Factor: 8.097].
26.	Sharika R, Subbiah P and **Balamurugan K**. (2018). Studies on reproductive stress caused by candidate Gram positive and Gram negative bacteria using model organism, Caenorhabditis elegans. Gene 649:113-126; https://doi.org/10.1016 /j.gene.2018.01.088; [Country: UK; Elsevier Ltd; Impact Factor: 2.415]
27.	Kavitha S, Pooranachithra M, **Balamurugan K** and Goel G (2018) Probiotic mediated colonization resistance against *E.coli* infection in experimentally challenged *C. elegans*. Microbial Pathogenesis. [Elsevier] [IF: 2.332] (In Press).
28.	Kamaladevi A, Marudhupandiyan S and **Balamurugan K** (2017). Model system based proteomics to understand the host response during bacterial infections. Molecular BioSystems 13: 2489-2497. DOI: 10.1039/C7MB00372B; [Country: UK; Royal Society of Chemistry] (Impact Factor: 2.781).
29.	Vigneshwari L and **Balamurugan, K**. Involvement of O-GlcNAcylation in *Caenorhabditis elegans* during pathogenic infection. FEBS Journal. Volume 75: Page 75. [ISSN: 1742-464X]; [Wiley; UK] (Impact Factor: 4.237);
30.	Kamaladevi A and Balamurugan K (2017). Global proteomics revealed *Klebsiella pneumoniae* induced autophagy and oxidative stress in *Caenorhabditis elegans* by inhibiting PI3K/AKT/mTOR pathway during infection. Frontiers in Cellular and Infection Microbiology 7:393; DOI: 10.3389/fcimb.2017.00393 [Country: Switzerland; Frontiers Media S. A.] (Impact Factor: 4.3).
31.	Marudhupandiyan S, Prithika U, Balasubramaniam B and **Balamurugan K** (2017). RACK-1, a

	multifaceted regulator is required for C. elegans innate immunity against S. flexneri M9OT infection. Developmental and Comparative Immunology. Vol. 74; September 2017, Pages 227-236. DOI:10.1016/j.dci.2017.05.008 [Country: UK; Elsevier] (Impact Factor: 3.62).
32.	Dhanashree, Sharika R, **Balamurugan K** and Rajagopal K (2017). Bifid shape is intrinsic to *Bifidobacterium adolescentis*. Front. Microbiol. 8:478. doi: 10.3389/ fmicb.2017.00478. (Impact Factor: 4.165).
33.	Prithika U, Vikneswari R and **Balamurugan K** (2016). Short term memory of *Caenorhabditis elegans* against bacterial pathogens involves CREB transcription factor. Immunobiology. DOI: 10.1016/j.imbio.2016.12.008 [Country: Netherlands; Publisher: Elsevier BV] (Impact Factor: 2.99).
34.	Marudhupandiyan S and **Balamurugan K** (2016). Intrinsic JNK-MAPK pathway involvement requires *daf-16* mediated immune response during *Shigella flexneri* infection in *C. elegans*". Immunologic Research DOI: 10.1007/s12026-016-8879-6. [Country: USA; Springer] (Impact Factor: 2.934).
35.	Kamaladevi A and **Balamurugan K** (2016). *Lactobacillus casei* triggers TLR mediated RACK-1 dependent p38 MAPK pathway *in Caenorhabditis elegans* to resist *Klebsiella pneumoniae* infection. Food & Function 7: 3211- 3223. DOI: 10.1039/C6FO00510A [Country: UK; Royal Society of Chemistry] (Impact Factor: 2.791).
36.	Prithika U, Deepa V and **Balamurugan K** (2016). External induction of heat shock stimulates the immune response and longevity of *C. elegans* towards pathogen exposure. Innate Immunity 22(6): 466-478. DOI: 10.1177/1753425916654557 [Country: UK; SAGE Publishing] (Impact Factor: 3.271)
37.	Vigneshkumar B, Durai S, Kundu S and **Balamurugan K** (2016). Proteome Analysis Reveals Translational Inhibition of *Caenorhabditis elegans* enhances susceptibility to *Pseudomonas aeruginosa* PAO1 pathogenesis. Journal of Proteomics 145: Pages 141-152. DOI: 10.1016/j.jprot.2016.03.047 [Elsevier, Country: UK] (Impact Factor: 3.888)
38.	JebaMercy G, Durai S, Prithika U, Marudhupandiyan S, Dasauni P, Kundu S and **Balamurugan K** (2016). Role of DAF-21 in *Caenorhabditis elegans* immunity against *Proteus mirabilis* infection. Journal of Proteomics 145: Pages 81-90. DOI:10.1016/j.jprot.2016.03.047 [Elsevier, Country: UK] (Impact Factor: 3.888)
39.	Kamaladevi A and **Balamurugan K** (2016). Lipopolysaccharide of *Klebsiella pneumoniae* attenuates immunity of *Caenorhabditis elegans* and evades by altering its supramolecular structure. *RSC Advances* 6:30070-30080. DOI: 10.1039/C5RA18206A. [Country: UK; Royal Society of Chemistry] (Impact Factor 3.84)
40.	Prasanth MI, Santoshram GS, Bhaskar JP and **Balamurugan K** (2016). Ultraviolet-A triggers photoaging in model nematode *Caenorhabditis elegans* in a DAF- 16 dependent pathway. *AGE* (Dordr) 38(27): 1-13; DOI: 10.1007/s11357-016-9889-y (Country: American Aging Association, Dordrecht, The Netherlands; Publisher: Springer; Impact Factor: 3.445)
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